playbook as shown below:

```
$ ansible-playbook -i inventory/kvm/inventory playbooks/
configuration/apache.yml -K
SUDO password:

PLAY [Install Apache web server] ****************************
TASK [setup] ************************************************
ok: [ubuntu]
TASK [Update the software package repository] ***************
changed: [ubuntu]
TASK [Install Apache] ***************************************
changed: [ubuntu] => (item=[u'apache2'])
TASK [wait_for] *********************************************
ok: [ubuntu]
PLAY RECAP **************************************************
ubuntu                    : ok=4    changed=2
unreachable=0    failed=0
```

The '-K' option is to prompt for the sudo password for the 'xetex' user. You can increase the level of verbosity in the Ansible output by passing '-vvvv' at the end of the *ansible-playbook* command. The more number of times 'v' occurs, the greater is the verbosity level.

If you now open *http://192.168.122.250*, you should be able to see the default Apache2 *index.html* page as shown in Figure 1.

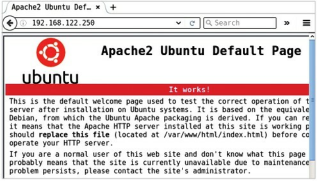

Figure 1: Apache2 Ubuntu default page

Installing MySQL

The next step is to install the MySQL database server. The corresponding playbook is provided below:

```
---
- name: Install MySQL database server
  hosts: ubuntu
  become: yes
  become_method: sudo
  gather_facts: true
  tags: [database]
  tasks:
    - name: Update the software package repository
      apt:
        update_cache: yes
    - name: Install MySQL
      package:
        name: "{{ item }}"
        state: latest
      with_items:
        - mysql-server
        - mysql-client
        - python-mysqldb
    - name: Start the server
      service:
        name: mysql
        state: started
    - wait_for:
        port: 3306
    - mysql_user:
        name: guest
        password: '*F7B659FE10CA9FAC576D358A16CC1BC646762FB2'
        encrypted: yes
        priv: '*.*:ALL,GRANT'
        state: present
```

The package repository is updated and the necessary MySQL packages are installed. The database server is then started, and we wait for the server to be up and running. A 'guest' user account with 'osfy' as the password is created for use in our Web application. The chosen password is just for demonstration purposes. When used in production, please select a strong password with special characters and numerals.

You can compute the hash for a password from the MySQL client, as shown below:

```
mysql> SELECT PASSWORD('osfy');
+-------------------------------------------+
| PASSWORD('osfy')                          |
+-------------------------------------------+
| *F7B659FE10CA9FAC576D358A16CC1BC646762FB2 |
+-------------------------------------------+
1 row in set (0.00 sec)
```

An execution run to install MySQL is as follows:

```
$ ansible-playbook -i inventory/kvm/inventory playbooks/
configuration/mysql.yml -K
SUDO password:
PLAY [Install MySQL database server] ************************
TASK [setup] ************************************************
ok: [ubuntu]
TASK [Update the software package repository] ***************
changed: [ubuntu]
TASK [Install MySQL] ****************************************
changed: [ubuntu] => (item=[u'mysql-server', u'mysql-client',
u'python-mysqldb'])
```